TruthLens: A Verification-Gated Pipeline for Evidence-Grounded Fact-Checking of Short-Form Political Video

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Abstract—Short-form video is now a primary vector for political misinformation, yet most automated fact-checking systems are evaluated on whether they reach a correct final label, not on whether the evidence chain that produced it is real, and few are evaluated against independent, held-out ground truth at all. We present TruthLens, a verification-gated pipeline for short-form political video (Instagram Reels) that decomposes a post into atomic claims, researches each against the open web, and subjects every LLM-generated verdict to a deterministic, non-LLM validator before publication – checking citation existence, source-fetch existence, numeric grounding, and, newly, label/reasoning consistency. We evaluate TruthLens on a held-out benchmark of 9 real Instagram posts independently fact-checked by professional Indian fact-checking organizations (BOOM Live, Alt News, Factly), disjoint from all development data, against three baselines that hold the underlying model and search infrastructure constant (LLM-only; single-shot search+LLM; search+RAG+LLM). A skeptical post-hoc audit of our own experimental design found that these baselines had, in every run across this project, been given a clean, human-written claim summary rather than the claim TruthLens's own extraction stage actually produced – an undisclosed confound that inflated baseline performance relative to our own system, contradicting our own stated methodology. We report this bug and its fix with the same prominence as any other finding. Corrected – baselines re-run per real extracted claim, aggregated with the identical deterministic rule TruthLens itself uses – full TruthLens (33.3%, 2/6) beats two of the three baselines outright (0.0% each) and trails the third by 16.7 points (down from a 33.4-point gap before the fix), on the paired six-item subset where every configuration produced a verdict. We further show that two general, ground-truth-independent validator fixes measurably improve validator recall (16.7%→40%, on the same 9 cases, 100% precision throughout) without moving reel-level accuracy at all – concrete evidence that verdict-label accuracy and what we formally define as support validity (whether a verdict's citations, sources, numbers, and label/reasoning consistency hold up against its own evidence matrix) are separable, independently-moving properties, which we argue is closer to this architecture's actual value proposition than a label-accuracy contest against simpler baselines. A separate counterfactual claim-decomposition reanalysis ($n = 4$) isolates deterministic aggregation as a real contributor (50.0% vs. 25.0% accuracy) once claim extraction has more than one claim to work with. We report the evidence-quality picture behind these results (a primary-source domain-restriction fix moving source-tier classification into government/legal tiers from ∼11% to 95% of retrieved sources; a distinct, still-open gap between finding topically-relevant primary sources (68.75%) and extracting usable evidence from them (18.75%); a bug found and fixed mid-evaluation in which 47.1% of real evidence rows carried an empty explanation field), a multimodal claim-coverage experiment in which OCR-derived evidence recovered a claim audio transcription missed entirely, and honestly scoped non-results for political-bias and confidence-calibration questions that this dataset's size cannot yet support. Every number in this paper is reported with its exact sample size and, where a proportion, a Wilson confidence interval; where an experiment could not be run at a scientifically meaningful scale, or where our own methodology was found to be flawed, we say so rather than force or hide it.

Index Terms—automated fact-checking, large language models, evidence grounding, retrieval-augmented generation, misinformation detection, short-form video, claim verification, source credibility, experimental methodology

I. Introduction

Political misinformation increasingly spreads through short-form video – Instagram Reels, YouTube Shorts, TikTok – rather than text [1]. This shift matters for automated fact-checking research in two ways. First, the input is multi-modal and lossy: a claim may be spoken (transcript), displayed as on-screen text (OCR), or only implied by a caption, and a system that treats a reel as a single block of text discards exactly the structure a fact-checker needs. Second, and more consequentially, the dominant evaluation paradigm for automated fact-checking – did the system output the correct veracity label, measured against a benchmark such as FEVER [2] or LIAR [3] – says nothing about whether the evidence a system cites in support of that label is real, or whether a correct label was reached by checking the actual claim being made rather than a different, tangential one. An LLM asked to fact-check a claim can produce a fluent, well-cited verdict whose citations point to sources it never retrieved, whose numbers never appeared in any source it read, or whose confident label contradicts its own stated reasoning.

We built TruthLens to make that failure mode structurally difficult rather than merely discouraged by prompting. TruthLens ingests a political short-form video (or photo post), decomposes it into atomic factual claims, researches each against the open web through a tiered-source retrieval process, and only then proposes a claim-level verdict – a hard evidence-before-verdict ordering enforced architecturally, not just by instruction (Section IV). Every LLM-generated verdict is additionally subject to a deterministic, non-LLM validation layer before it can be published, and every pipeline stage escalates from a free, locally hosted model to a paid one only when a deterministic quality gate specific to that stage's own failure modes fails.

This paper reports the first evaluation of TruthLens against independent, held-out ground truth, rather than development-time telemetry alone. We curated a 9-item benchmark of real Instagram posts, each independently fact-checked by a professional Indian fact-checking organization (BOOM Live, Alt News, Factly), disjoint from every post used during development (Section VI), and compared TruthLens against three baselines that hold the underlying model and search infrastructure constant – isolating what TruthLens's specific multi-stage architecture adds beyond raw search access (Section V). We report every result this comparison produced, including one that is not flattering, but not about TruthLens: a subsequent, skeptical audit of our own experimental design found that the baselines had been given a cleaner claim input than TruthLens itself ever worked with, contradicting our own stated methodology, undetected across the entire program until this audit (Section VII-A). We report the bug and its fix with the same prominence as any other finding, because a flaw that inflates a comparison against our own system is exactly the kind of result our own protocol requires us to surface rather than quietly correct. Once corrected, full TruthLens beats two of the three baselines outright and trails the third by a substantially smaller margin than the uncorrected comparison showed (Section VII) – and a companion counterfactual analysis of claim decomposition shows the same architecture's aggregation mechanism demonstrably helps once claim extraction has more than one claim to work with (Section VIII). We treat the correction, the corrected result, and the ablation as more scientifically useful, together, than any single headline number would have been on its own.

This work is organized around six research questions, matching TruthLens's own architectural claims one-to-one so that each is either supported by a specific experiment or explicitly withdrawn: whether deterministic verification reduces unsupported output (RQ1); whether multi-stage decomposition and retrieval beat single-shot search (RQ2); whether multimodal extraction improves coverage of non-textual misinformation (RQ3); whether source-tiering improves evidence quality (RQ4); whether the system maintains comparable standards across political actors (RQ5); and whether its stated confidence correlates with correctness (RQ6). Section III states these formally alongside this paper's specific, experimentally-supported contributions.

We are explicit throughout about the evidentiary status of every claim: 9 held-out items (6 with a resolved verdict from every configuration), single-annotator Tier-2 judgment where used, and RQ5/RQ6 explicitly deferred rather than forced past what this sample size can support (Section XI). Sections XIII and XV state exactly what would need to change before any number here should be read as a stable, generalizable estimate.

II. Related Work

Automated fact-checking and LLM factuality evaluation have both grown into large enough literatures to warrant their own surveys [4], [5]; we focus below on the specific sub-areas TruthLens draws on and departs from, rather than attempting a comprehensive survey ourselves. None of the ideas below – retrieval-augmented LLM fact-checking, claim decomposition, source-tiering, or multimodal misinformation detection – originate with this work; our contribution, argued for in Section III and evaluated in Sections VII–XI, is a specific combination of them gated by deterministic verification, evaluated honestly including where it does not yet outperform simpler alternatives.

A. Claim detection and decomposition

Claim detection for fact-checking spans monolingual, multilingual, and cross-lingual settings [6], all of which are relevant to TruthLens's mixed English/Hindi/Urdu input. ClaimBuster [7] established check-worthy claim detection as a distinct pipeline stage, using a TF-IDF/POS/NER-featured SVM to rank sentences from political debate transcripts by how worth fact-checking they are. TruthLens's claim-extraction stage performs a related but distinct task: rather than ranking existing sentences, an LLM decomposes a transcript, OCR stream, and caption into new atomic claims, splitting compound statements into independently checkable parts, following the general claim-decomposition strategy used in systems such as ProgramFC, which decomposes complex claims into sub-questions before per-question retrieval. We extend this with a grounding requirement specific to video: each extracted claim's source_quote, when present, must be a verbatim substring of the actual transcript/OCR text, checked deterministically, not merely asserted by the model. Section VIII reports the first controlled measurement, on real held-out data, of whether this decomposition step actually improves downstream verdict correctness relative to researching only the single highest-importance claim.

B. Fact-checking benchmarks

FEVER [2] (Wikipedia-derived claims, Supports/ Refutes/NotEnoughInfo) and LIAR [3] (12.8K PolitiFact statements, six-way truthfulness) are the dominant benchmarks for claim-verification models, and both evaluate a system's final label against ground truth without examining whether the evidence chain that produced the label is real. TruthLens's own benchmark (Section VI) is a first, deliberately modest step toward the same kind of independent, held-out evaluation for our domain – 9 Tier-1 items, three orders of magnitude smaller than FEVER or LIAR, sourced from real Indian professional fact-checks of Instagram posts rather than curated from Wikipedia or PolitiFact, and not yet at a scale that supports the same statistical claims. Reaching that scale, at the sourcing hit-rate this domain actually supports (∼8%, Section VI), remains the single most important item of future work (Section XV).

C. LLM hallucination and grounding verification

A growing body of work targets citation and grounding hallucination in LLM output specifically. A preregistered, hand-scored evaluation of leading commercial legal AI research tools found each hallucinates on 17–33% of queries [8], and recent systems such as CiteCheck [9] detect hallucinated references by grounding each citation against external sources with a structured LLM verifier. Our validation layer (Section IV-D1) shares the goal but differs in mechanism: rather than using a second LLM call to judge whether a citation is plausible, TruthLens's checks are deterministic set-membership and substring tests against data the pipeline itself already fetched (does the cited evidence_id exist; was the cited source actually retrieved; does the asserted number appear in the cited passage's stored text; does the reasoning's own stated conclusion match its label). This trades generality (we cannot catch fabricated content that happens to reuse a real, cited evidence ID incorrectly, or reasoning phrased differently from the phrasings we have observed and checked for – Section IX discloses this latter limitation as a concrete, measured threat to validity, not a hypothetical one) for auditability and zero marginal LLM cost per check. A recent analysis of citation-grounding oracles in the legal domain is a useful caution for exactly this kind of check: it shows that an apparent hallucination rate measured by verifying citations against a reference graph is highly sensitive to that graph's coverage, not just to the model being measured, with the same responses scoring 15–21% "hallucinated" against a sparse snapshot of a citation registry and under 1.1% against a denser one built from the same source [10]. TruthLens's own number-grounding check has an analogous blind spot: it can only confirm a number against sources the pipeline itself fetched, so it cannot distinguish "this number is fabricated" from "this number is real but supported only by a source we did not retrieve." We flag this as an open limitation rather than a solved problem, and follow Rule 8 of our own evaluation protocol throughout in not calling any of these rates a "hallucination rate" without human annotation behind it – we use grounding-constraint violation rate for what the deterministic validator itself checks, and unsupported generation rate for the human-judged quantity in Section IX.

D. Political fact-checking with LLMs

Most directly relevant to TruthLens's domain, DeVerna et al. [11] evaluated 15 LLMs against over 6,000 PolitiFact-verified claims and found that reasoning ability provided "minimal benefit" and raw web search only "moderate gains" for political fact-checking accuracy, while providing curated, high-quality context improved macro-F1 by an average of 233% across models. This is strong independent evidence for TruthLens's core design bet: that pipeline-level evidence curation (multi-query research planning, source tiering, per-source relevance filtering) matters more than which LLM does the final reasoning step. Our own headline result (Section VII) complicates a naive reading of this at small n rather than contradicting it: TruthLens's curation-heavy architecture underperformed a single-shot search baseline on our 6-item paired sample, but the deficit traces to claim-extraction coverage failures upstream of curation (two items with zero verifiable claims extracted at all) and two validator false negatives, not to curation itself failing when it runs – and our claim-decomposition ablation (Section VIII) shows a curation-adjacent mechanism (decomposition plus deterministic aggregation) delivering a real accuracy gain once it has claims to work with. We view TruthLens as, in part, an applied instantiation and extension of DeVerna et al.'s finding to a harder, multi-modal input domain, with this paper's contribution being an honest account of where that instantiation currently succeeds and where it does not.

E. Short-form video misinformation

TikTec [1] detects misleading COVID-19 short videos on TikTok by jointly modeling visual, audio, and textual signal, using captions to focus attention on the most informative frames. Recent work on checkworthiness detection specifically targets multilingual short-form video [12]. TruthLens differs from this line of work in task framing: rather than a binary/graded misinformation classifier over the video as a whole, TruthLens performs open-domain, claim-level, cited verification against live web evidence, closer in spirit to FEVER-style verification than to TikTec-style classification, but applied to video-derived claims rather than Wikipedia sentences. Section X reports a structural finding relevant to this whole line of work: in the majority of our Tier-1 sample, the false or misleading claim does not appear anywhere in the specific post's own transcript, caption, or on-screen text at all – consistent with genuine footage being recirculated under a new, false caption elsewhere, a pattern neither TikTec-style classification nor TruthLens's own text-derived claim extraction currently addresses.

F. Political neutrality of automated fact-checkers

LLMs are not neutral instruments; recent evaluations document measurable partisan asymmetries in frontier models under politically charged prompts [13]. Because a visibly biased automated fact-checker is worse than none for a politically sensitive application, every TruthLens system prompt includes an explicit neutrality clause instructing the model to apply identical evidentiary standards regardless of which political actor is named. We attempted a quantitative bias evaluation as RQ5 of this program and report, in Section XI, exactly why it could not be run at a meaningful sample size rather than either skip it silently or force a result our data cannot support.

G. Cost-aware model cascading

FrugalGPT [14] demonstrated that routing most queries to a cheap model and escalating only hard cases to an expensive one can sharply cut inference cost with little accuracy loss. TruthLens applies the same principle in reverse-priority form suited to a zero-marginal-cost constraint: every stage defaults to a free, locally hosted model (Llama 3.2 via Ollama) rather than a cheap paid one, and escalates to a paid model (Gemini) only when a deterministic quality gate – not a learned cost/accuracy predictor as in FrugalGPT – detects that the local model's output is schema-valid but substantively deficient in a way specific to that stage (Section IV-D3).

H. Positioning against related systems

Table I summarizes where TruthLens sits relative to the systems discussed above, on dimensions each subsection has already individually justified. We include only systems whose relevant capabilities are established by the citations already discussed in this section, to avoid characterizing a system's design beyond what we have verified. TruthLens is the only row combining multimodal input, claim decomposition, source tiering, deterministic (not LLM-judged) output verification, and held-out professional fact-check ground truth – combining these is TruthLens's contribution, as Section III states explicitly; no individual capability in this table originates with TruthLens.

Stated plainly, per this paper's own governing discipline against overclaiming: TruthLens does not claim novelty for any single column in Table I. Multimodal short-form-video analysis predates it (TikTec); political-domain fact-checking with LLM-mediated retrieval predates it (DeVerna et al.); citation grounding against an external source predates it, including via a non-deterministic LLM -judged mechanism this paper argues against on cost and auditability grounds, not on the underlying goal (CiteCheck). What we have not found in the literature we reviewed is a system combining held-out professional-fact-check evaluation, multimodal claim extraction, source -tiered retrieval, and deterministic (non-LLM) output verification together, evaluated on real, recirculated short-form political video – which is the specific gap Sections IV–XI target and evaluate honestly, including where the combination does not yet outperform a simpler alternative (Section VII).

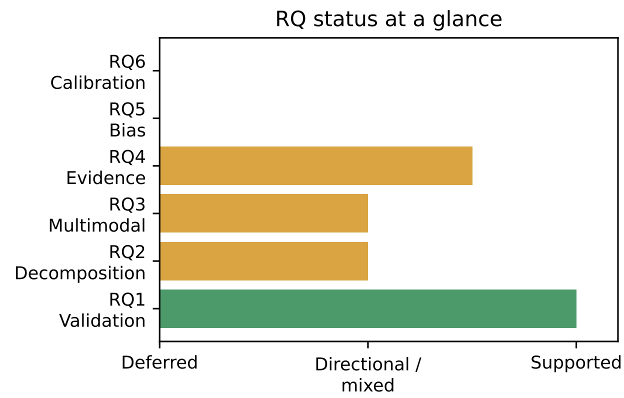

Fig. 1. Status of each research question at a glance (green = supported, amber = directional/mixed, red = deferred). Derived from Table II; not new data.

III. Research Questions and Contributions

Table II states the six research questions this program was designed against, where each is answered, and its evidentiary status, following Rule 12 of our own governing protocol: an RQ that cannot be evaluated at a meaningful sample size is reported as deferred, not silently dropped or forced.

This paper makes the following contributions, limited to what the experiments in Sections VII–XI actually support:

1) A held-out, Tier-1 evaluation benchmark and controlled baseline/counterfactual comparison (Sections V– VI), the first evaluation of TruthLens against independent professional ground truth rather than development-time telemetry, holding model, search infrastructure, and (Section VII-A) claim -extraction input constant across every configuration so that measured differences in Section VII are attributable to downstream architecture specifically, not to which configuration had an easier claim to reason about.
2) A documented, fixed methodological confound in our own baseline design, reported with the same prominence as any other finding. A skeptical post-hoc audit of this program's own experimental design (Section VII-A) found that Baselines 1–3 had, in every run across this project, been given a clean, human-written claim summary rather than TruthLens's own extracted claim, contradicting our own stated methodology and inflating baseline performance relative to our own system. We fixed it and re-ran the comparison rather than silently correcting the number or leaving it undisclosed, reversing this program's headline result in the process.
3) An empirical demonstration that verdict-label accuracy and what we formally define as support validity are separable, independently-moving properties (Section IX). Two general, ground-truth-

TABLE I
Positioning against related systems (✓ = present, ✗ = absent, ? = not established by the citation discussed)

System	Multimodal	Short-video	Political	Claim decomp.	Retrieval/RAG	Source tiering	Verification mechanism
FEVER [2]	✗	✗	✗	✗	✓	✗	none (label-only benchmark)
LIAR [3]	✗	✗	✓	✗	✗	✗	none (label-only benchmark)
ClaimBuster [7]	✗	✗	✓	✗	✗	✗	n/a (checkworthiness only)
TikTec [1]	✓	✓	✗	✗	✗	✗	none (classifier output)
DeVerna et al. [11]	✗	✗	✓	✗	✓	✗	none (studies LLM judges)
CiteCheck [9]	✗	✗	✗	✗	?	✗	LLM-judged citation check
TruthLens (this paper)	✓	✓	✓	✓	✓	✓	deterministic, 4-check

TABLE II
Research questions, where answered, and status

RQ	Question	Status
1	Does deterministic post-hoc validation reduce unsupported verdict output?	Yes, directional (§IX, $n = 9$)
2	Does multi-stage decomposition + retrieval beat single-shot search+LLM?	Partially, once claim input is corrected: beats 2 of 3 baselines, trails the 3rd (§VII); yes for aggregation specifically (§VIII, $n = 4$)
3	Does multimodal extraction improve coverage of non-textual misinformation?	Directional yes (§X, $n = 6$)
4	Does source-tiering improve evidence quality?	Yes for tier classification; gap remains to usable evidence (§IX)
5	Comparable standards across political actors?	Deferred – no matched pairs exist (§XI)
6	Does confidence correlate with correctness?	Deferred – sample too small to calibrate (§XI)

independent validator fixes measurably improved validator recall (16.7%→40%, $n = 9$, 100% precision throughout) while leaving reel-level accuracy completely unchanged (2/6 before and after) – a concrete case where a system's outputs became measurably more supported by their own cited evidence without becoming more "accurate" by the bucket-matching metric standard benchmarks use.

4) A claim-decomposition counterfactual analysis isolating the clearest component-level result in this program where TruthLens's aggregation mechanism demonstrably outperformed a simpler alternative (50.0% vs. 25.0% accuracy, $n = 4$, Section VIII).
5) A four-way evidence-quality analysis that separates source -tier classification, topical relevance, fetch success, and usable -evidence extraction into four distinct, independently reported metrics (Section IX), showing the system's retrieval finds real, on-topic primary sources at a meaningfully higher rate (68.75%) than its own usable-evidence rate (18.75%) would suggest – the bottleneck is source specificity, not retrieval relevance.
6) A cross-referenced taxonomy of real failure modes (Section XII), each found only by running the complete system against real content end-to-end and each independently fixed with a deterministic check and a regression test built from the exact real case that surfaced it, offered as a practically-oriented complement to the benchmark numbers above.

TruthLens does not claim to have invented retrieval-augmented LLM fact-checking, claim decomposition, source-credibility tiering, or multimodal misinformation detection – each independently pre-dates this work, as Section II discusses. Its contribution is the specific combination of these mechanisms gated by deterministic, non-LLM verification, evaluated against independent ground truth including where that combination does not yet outperform a simpler alternative.

IV. TruthLens Architecture

TruthLens is a FastAPI backend with a PostgreSQL/pgvector datastore, a Next.js review dashboard, and an object store (S3-compatible) for media and rendered output. The pipeline runs in the following strict order for every reel, and no stage may begin before the stage before it has persisted its output to the database:

1) Ingestion. The source video is downloaded (or manually uploaded), audio is extracted and transcribed, on-screen text is sampled and OCR'd at a fixed interval, and a representative thumbnail frame is selected. A single-image (photo) post follows a parallel path with no video stream at all (Section IV-C).
2) Claim extraction. An LLM decomposes the transcript, OCR text, and caption into atomic claims, each classified as factual, opinion, prediction, satire, or rhetorical, with only factual claims specific enough to research marked verifiable.
3) Research planning. For each verifiable claim, an LLM proposes 2–5 targeted search queries, explicitly instructed to include at least one query aimed at primary sources (official/government domains) when plausible.
4) Search and fetch. Each query is run against a search provider; result pages are fetched and their full text extracted (not just the search snippet).

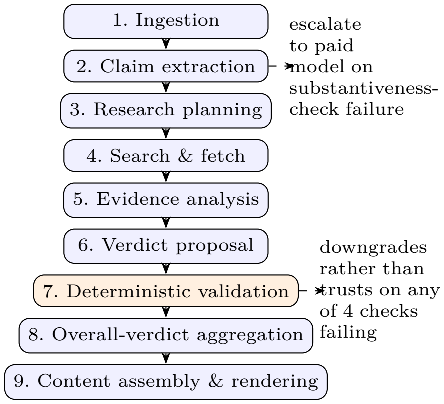

Fig. 2. The 9-stage TruthLens pipeline (Section IV). No stage begins before the one before it has persisted its output. Stage 7 (orange) is the deterministic, non-LLM validation gate (Section IV-D1); five stages (2, 5, 6, and the two content -generation calls inside stage 9, plus stage 5's newer sibling check) run their own substantiveness check before accepting local-model output (Section IV-D3).

5) Evidence analysis. For each fetched source, an LLM judges stance (supports/contradicts/provides_ context/irrelevant) against the claim, grounded in that source's actual extracted text only.
6) Verdict proposal. Given the claim and its full evidence matrix (every evidence row, with source reliability and stance), an LLM proposes a verdict label, confidence, and cited evidence IDs.
7) Deterministic validation (Section IV-D1).
8) Overall-verdict aggregation. Once every claim covered by a fact-check has a validated verdict, a fixed rule table – not an LLM call – derives one reel-level verdict.
9) Content assembly and rendering. A carousel of slides (headline poster, claims/quote, evidence, conclusion) is assembled, drawing almost all content directly from already-validated database rows; only headline phrasing and a short "why" paragraph pass through one additional narrow, schema-constrained LLM call each, both with their own grounding checks (Section IV-D3).

A. Source tiering

Every retrieved source is scored along an 8-dimension weighted reliability rubric and assigned to one of eight tiers: primary_government, primary_legal, primary_data, news_wire, established_news, academic, factcheck_org, or other. Tier membership determines both which sources are eligible to be surfaced as a claim's "primary source" citation on the rendered carousel and, in aggregate, the reliability-weighted confidence of the evidence matrix handed to the verdict-proposal stage. RQ4 (Section IX) evaluates this mechanism directly. The eight dimensions' weights (primary-source status and publication reputation each 0.20; directness and corroboration each 0.15; evidence transparency and recency each 0.10; author identity and conflict of interest each 0.05) are hand-set manual priors, not derived from any formal method, learned fit, or literature citation – and, verified directly against this file's git history for this audit, unmodified since the first commit of this project, before any evaluation dataset existed. We state this plainly rather than imply an empirical grounding these weights do not have; a sensitivity analysis comparing this weighting against an unweighted rubric and at least one alternative reasonable weighting, to check whether final verdicts are sensitive to the specific values chosen, is unperformed and named in Section XV.

B. Multi-claim, reel-level verdicts

TruthLens builds one fact-check per reel, covering every verifiable claim extracted from it, because a single short-form video routinely makes several claims of differing veracity, and collapsing them into one per-claim verdict discards exactly the "partially true, partially misleading" structure a reader needs. A fixed rule table governs how per-claim verdicts combine into the one verdict actually shown on the published carousel (Section IV-D4); Section VIII reports the first controlled test of whether this multi-claim structure itself improves accuracy over researching only the single highest -importance claim.

C. Multi-modal ingestion: video and photo posts

Not every short-form political post is a video. TruthLens handles both: when the video downloader reports no video stream at all, the pipeline falls back to the post's Open Graph metadata (image URL, caption, engagement counts) fetched directly from the page rather than through the video-download tool, which does not expose a photo-only code path. A media_type field (video/photo) drives branching at exactly one point downstream: audio extraction and transcription are skipped for a photo (there is no audio), but OCR and vision-context analysis needed no new code at all, because both already operated on a plain list of image paths rather than assuming a video – a single photo is a one-element case of the same interface a video's sampled frames already satisfy. Section X evaluates claim coverage across this system's three real input-modality conditions (text-only, +OCR, +OCR+vision) on real ingested held-out data.

D. Making the model's output more supportable, not just more capable

Across this project we have found that improving TruthLens's output is substantially a question about the pipeline around the model rather than the model itself. We make this precise with a single formal construct used throughout the rest of this paper in place of the vaguer word "trustworthy": a verdict has support validity to the extent that every specific, checkable factual assertion embedded in it – its cited evidence IDs, the sources those citations point to, the numbers stated in its reasoning, and the logical relationship between its stated reasoning and its chosen label – can be verified against data already present in the pipeline's own evidence matrix, without invoking a second, equally fallible model judgment. Support validity is deliberately narrower than, and does not imply, either label accuracy (Section VII) or general absence of hallucination: a verdict can have perfect support validity by this definition and still be wrong, if the model correctly cited and correctly quoted evidence it nonetheless misinterpreted, or if retrieval never found the right evidence in the first place (Section IX's four-way evidence metric addresses that separate question). This subsection describes the mechanisms we built to make support validity checkable; Sections VII–XI evaluate whether, and how well, each actually works.

1) Deterministic evidence-grounding validation: After the verdict-proposal LLM call returns a structured VerdictProposal (label, confidence, reasoning, cited evidence IDs), TruthLens runs four checks against data already in the database, none of which invoke an LLM:

- Citation existence: every ID in cited_evidence_ ids must correspond to an evidence row that was actually passed to the model in its evidence matrix.
- Source-fetch existence: every cited evidence row's source must have a stored, successfully fetched full-text passage – a citation cannot point at a source the pipeline only planned to fetch.
- Number grounding: every numeral appearing in the model's reasoning_summary is extracted and checked for appearance in at least one cited source's stored passage text. UUID-shaped hex sequences (from internal citation markup) are stripped before number extraction – a real bug, described in Section XII, in which digits from a cited evidence ID's own UUID were being extracted as apparent "unsupported statistics."
- Reasoning/label consistency (new). A verdict whose own reasoning_summary states that no reliable evidence was found (matched against a fixed phrase list, e.g. "does not provide any reliable," "no reliable sources") but whose label is a confident value other than UNVERIFIED is downgraded to UNVERIFIED. Added in direct response to Section IX's finding that this specific pattern was the largest single category of validator false negative; its circularity as a threat to validity is disclosed, not hidden, in Section IX.

These four checks are the entire current operationalization of support validity – deliberately not a proxy for evidence-to-claim topical relevance (Section IX's job), evidence-to -reasoning stance correctness (currently human-audited only, not deterministically checked at all – a named gap, not an oversight), or entity consistency (not yet implemented; a concrete, scoped item in Section XV). A verdict that fails any check is not discarded – it is downgraded: persisted with a validation_status other than passed, an appended, clearly delimited [VALIDATION NOTE: ...] explaining what failed, and excluded from citation lists shown to a reader. This preserves the model's underlying verdict label (still potentially useful to a human reviewer) while refusing to let its specific, unverified justification reach a reader unflagged.

2) The RESEARCH_FAILED / UNVERIFIED distinction: UNVERIFIED is a legitimate, informative output: it means the system genuinely searched and found insufficient evidence to support or refute a claim. A distinct RESEARCH_FAILED claim status is raised only when all queries for a claim fail at the infrastructure level (not when queries succeed but return no relevant results), and a publication gate refuses to build a fact-check whose primary claim is RESEARCH_FAILED. This is a direct, domain-specific instance of the broader selective-prediction principle that a system should distinguish "I don't know" from "I could not check."

3) Verification-gated model cascading: Every pipeline stage defaults to a free, locally hosted model. Five stages – claim extraction, evidence analysis, vision-context analysis, content generation (headline/why-paragraph), and verdict proposal – additionally run a deterministic substantiveness check on the local model's output before accepting it:

- Claim extraction, grounding: do at least half of the extracted verifiable claims have a source_quote that is an actual substring of the combined transcript/OCR/caption text?
- Claim extraction, substantiveness: does at least one extracted claim have non-empty text? An extraction with zero claims at all is not treated as a failure by this check – a genuine "found nothing" is a legitimate outcome, not a quality problem to retry.
- Evidence analysis, substantiveness (new). Does the per-source explanation field contain non-empty text? Added after Section IX found 47.1% (32 of 68) of real evidence rows from a live pipeline run carried an empty explanation – a stance label with no auditable reason behind it. Unlike the other checks in this list, this one fires once per retrieved source, not once per claim, since this is the one stage that calls the LLM per source.
- Vision-context, prompt-echo detection (new). Does the scene_description's own vocabulary overlap substantially with the vision system prompt's vocabulary, rather than describing new visual content? Added after this exact pattern – the local vision model echoing fragments of its own instructions back as if they were a scene description – recurred across at least four separate real reels (Section X). Calibrated against every real garbled and real coherent example collected this project: garbled outputs scored 0.26–0.29 on a word-overlap-with-prompt ratio; coherent descriptions scored 0.00–0.12; the threshold is set at

0.20, in the gap between them.

- Content generation / verdict reasoning: after stripping internal citation markup, does the remaining prose contain at least six real words? This check exists because a locally hosted model would sometimes produce a reasoning_summary composed entirely of citation markup with no prose at all – schema-valid, but explaining nothing to a reader.

On failure, the same call is retried once against a stronger paid model (Gemini) with an identical prompt, and the retry (whichever model produced it) is logged. This differs from a classic cascade [14] in that the escalation trigger is a deterministic grounding/substantiveness predicate specific to that stage's own observed failure modes, not a learned or generic confidence estimate. Illustrative development-time telemetry (predating the frozen held-out evaluation reported from Section V onward, and not part of it) found this gate firing on 3 of 7 claim-extraction completions, 2 of 5 content -generation completions, and 2 of 8 verdict-proposal completions – small, non-independent counts, but large enough to show the gate firing routinely rather than rarely.

4) Deterministic overall-verdict aggregation: Once every claim covered by a fact-check has a verdict, TruthLens derives one reel-level verdict via a fixed rule table over the set of per-claim verdict labels (weighted by whether a claim is "major," importance ≥ 0.5) – for example, all claims TRUE yields overall TRUE; a mix containing both TRUE/ MOSTLY_TRUE and FALSE yields MISLEADING. This is a plain combinatorial function, not a further LLM call, so the single highest-stakes number in the product – the verdict shown on the published carousel's first slide – remains fully auditable from already-validated per-claim verdicts rather than resting on a second, opaque judgment. Section VIII evaluates this mechanism directly.

5) Grounded corrections: what the evidence shows instead: A verdict of FALSE or UNVERIFIED alone is a weak deliverable – professional fact-checkers do not stop at "this is false"; they say what is actually true, with a source. The verdict-proposal call carries two optional fields, corrected_fact and context_note, populated from the same evidence matrix already given to that stage and held to the identical "nothing not already in the evidence" rule as reasoning_summary. corrected_fact is the specific alternative fact the evidence establishes (a different number, a different date), left null rather than guessed when no such alternative is established; context_note is background the evidence itself states, never the model's own speculation. We deliberately did not build the natural next request – speculation about downstream consequences of a claim – because that is forward-looking generation with no evidence to ground it against, exactly the failure category Section IV-D1's architecture exists to prevent. Both fields are validated independently of reasoning_summary and, whenever the parent verdict fails validation, neither field is trusted for display even if it individually passed its own check.

V. Experimental Methodology

A. Central research question and design

The central question motivating this evaluation: can a verification-gated, evidence-grounded architecture reduce unsupported fact-checking outputs while preserving or improving factual correctness in short-form political media? We answer this via a controlled comparison across five system configurations, all run against the same underlying local model (llama3.2 via Ollama) and, where the configuration has search access at all, the same search provider (DuckDuckGo) TruthLens itself uses in production – holding model and search-infrastructure quality constant so that a measured difference is attributable to architecture, not to a cheap-model-vs.-expensive-model confound.

B. System configurations

1) LLM-only (Baseline 1). Claim text $\rightarrow$ llama3.2 $\rightarrow$ verdict. No search, no retrieved evidence, no validation.
2) Search + LLM (Baseline 2). Claim $\rightarrow$ one DuckDuckGo search (top-5 title/snippet pairs, no full-page fetch) $\rightarrow$ a single LLM call $\rightarrow$ verdict. The claim text is passed to search verbatim – no research-planning stage – the single most important architectural difference from Baseline 3 and from TruthLens itself.
3) Search + RAG + LLM (Baseline 3). Claim $\rightarrow$ search $\rightarrow$ full-page fetch and text extraction for each result (reusing TruthLens's own fetch logic) $\rightarrow$ a single LLM call over the concatenated passages $\rightarrow$ verdict. Differs from Baseline 2 only in retrieval depth, isolating that variable from multi-stage decomposition and per-source evidence analysis, neither of which either baseline performs.
4) TruthLens minus validation (Baseline 4). The real pipeline, unmodified, except that validate_verdict()'s outcome is recorded but never applied to what is persisted – implemented as a runtime configuration flag (SKIP_VALIDATION) rather than a forked codepath, so this configuration can never silently drift from the real system. This is the primary ablation for RQ1.
5) Full TruthLens. The system exactly as it exists at the frozen evaluation tag, unmodified.

Claim decomposition, source tiering, per-source evidence analysis, and deterministic validation are each present only in the full system (and, for validation, deliberately absent in Baseline 4) – Baselines 1–3 operate on the claim as already extracted by TruthLens's own claim-extraction stage, since evaluating claim extraction itself is RQ3's job, not RQ2's. A real infrastructure bug was found and fixed while building Baselines 2–3: the first working draft called the same provider -factory function every production pipeline stage uses, which silently returns a Gemini-fallback-wrapped provider whenever an API key is configured in the environment – giving both baselines the same failure-rescue mechanism TruthLens's own architecture was being evaluated for having. Fixed by importing the local-only provider directly in both baseline scripts, caught during smoke testing before any number was reported anywhere (Section XII).

C. Ablations

Beyond Baseline 4 (the validation ablation, RQ1), two further ablations are in scope: a source-tier ablation (domain-restricted vs. unrestricted retrieval, Section IX) and a claim-decomposition ablation (single-claim-per-reel vs. full multi-claim decomposition, Section VIII), the latter computed from already-collected verdict data at zero additional LLM cost.

D. Metrics and statistical conventions

Every reported proportion states its exact numerator, denominator, and a 95% Wilson score confidence interval – chosen because it remains well-behaved at the small sample sizes this evaluation reaches, unlike a normal-approximation interval. Paired comparisons between two configurations on the identical item set use McNemar's exact test, reported alongside the raw contingency table, never a bare p-value. Accuracy is computed over a fixed bucketing of VerdictLabel onto ground truth (TRUE/MOSTLY_TRUE, FALSE/MOSTLY_FALSE, and each of MISLEADING/MISSING_CONTEXT/OUTDATED as their own bucket); UNVERIFIED is never counted as a match to any ground-truth label, including a ground truth that is itself contested or ambiguous – an abstention is not a correct answer by construction. Full formulas for every metric used in this paper, including the evidence-quality and validator metrics used in Section IX, are fixed in advance in the project's METRICS.md and are not altered after seeing results.

E. Held-out discipline

The evaluation dataset (Section VI) is checked item-by-item against already-ingested development content and excluded on any match. No prompt, threshold, or model choice was changed based on this dataset's contents or TruthLens's performance on it from the moment an item was added onward; the two general fixes reported in Section IX are deliberately ground-truth-independent – pure functions of the model's own output, motivated by a general principle rather than by making a specific held-out item score better – and this distinction, along with its limits, is argued explicitly rather than asserted (Section IX). The dataset was frozen at a tagged commit before Baselines 1–3 were run, and the full system's own code was frozen at a second tag before its held-out run; any code change made after either freeze is logged, not silently applied.

F. Honest scope reductions

This program deliberately targets less than an ideal-scale evaluation, and states exactly where and why, rather than presenting a smaller study as if it were a larger one:

- Dataset size: 9 items, not 50–100. A pilot check of 26 professional fact-check articles found a ∼8% hit rate for a usable, still-live, Instagram-embedding Tier-1 candidate; reaching even 20–30 items would require checking on the order of 250–375 more articles, a large but bounded manual task not completed within this program's window (Section VI).
- Single primary annotator, no inter-annotator agreement statistic. Tier-1 ground truth is a professional organization's own published verdict, authored by no one on this project; where a Tier-2 (single-annotator) judgment is used at all (draft claim-coverage and evidence-relevance labels only, never a headline verdict), it is explicitly marked llm_assisted_draft or single-annotator and not presented as adjudicated ground truth.
- RQ3, RQ5, and RQ6 are reported as directional or deferred, not as generalizable rates, because the sample size does not support a stronger claim (Sections X, XI).
- No p-values from tests that assume a sample size this program does not reach. Independent-samples t-tests and χ^2 tests are not used; Wilson intervals and McNemar's exact test are, and are reported as descriptions of uncertainty at this sample size, not claims of statistical significance.

VI. Dataset and Annotation

A. Ground-truth tiers

Tier 1 (used for every item in this evaluation): the claim in the Instagram post has already been verdicted by an independent professional fact-checking organization, and that organization's article demonstrably references or embeds this specific post, not merely a similar claim elsewhere. Ground truth is their published verdict, with the article URL stored for auditability. Tier 2 (specified, not used for any headline verdict in this evaluation): no professional fact-check exists; a project annotator labels directly from primary sources under a documented protocol. No item is ever labeled by TruthLens itself, or by any LLM, and then used as ground truth.

B. The 9-item benchmark

Table III lists every item. All 9 are Tier 1, sourced from BOOM Live, Alt News, and Factly. item-0003 (a CJP-investigated, AI-generated visual claim) could not be ingested after three independent attempts across this program (a persistent Instagram-side "empty media response"), and is excluded from every table in this paper, not scored as a failure.

TABLE III
Held-out benchmark: 9 Tier-1 items (all sourced independently of TruthLens or its developer)

ID	Ground truth	Source	Political actor	Claim type
item-0001	FALSE	BOOM Live	BJP	provenance
item-0002	FALSE	Alt News	Karni Sena	provenance (visual)
item-0003†	FALSE	BOOM Live	CJP	visual (AI-generated)
item-0004	TRUE	BOOM Live	Delhi Police (government)	provenance
item-0005	MISLEADING	Factly	Congress	provenance
item-0006	FALSE	BOOM Live	none (disaster misinfo)	event (visual)
item-0007	FALSE	Alt News	BJP	misleading context
item-0008	FALSE	Alt News	AAP	quote
item-0009	FALSE	Alt News	BJP	provenance

†Never successfully ingested; excluded from all analyses.

C. Sourcing method and hit rate

Items were sourced by searching professional fact-checking outlets for articles that embed a still-live Instagram post, prioritizing coverage gaps in claim type, language, and political actor as the set grew across three sourcing passes. The method's real, load-bearing constraint is a ∼8% hit rate (2 of the first 26 fact-check articles checked yielded a usable, still-live, directly-embedded Instagram post) – most professional fact-checks either do not embed the specific post, or the post has since been taken down. Every accepted item's content hash is checked against already-ingested development data and excluded on any match.

D. Inter-annotator agreement

No IAA statistic is computed, and none is fabricated to fill this gap. Tier-1 ground truth does not have an "annotator" in the IAA sense – the label is a single organization's single published verdict, and introducing a second in-project annotator to re-judge it would defeat the purpose of using independent ground truth at all. No Tier-2 items exist in the frozen dataset, so the condition under which IAA would be meaningful (two independent annotators judging the same items) does not currently arise. If Tier-2 items are added with only one available annotator, this will continue to be reported honestly as "single-annotator ground truth, no IAA computed," not papered over with an unearned kappa statistic.

E. Composition gaps, disclosed

Five of 9 items are provenance-type claims; statistic, law_policy, historical, and true_claim remain unrepresented after three sourcing passes – a plausible, though unproven, property of which claim types concentrate on Instagram specifically versus other platforms for this domain. Seven of 9 items are FALSE, likely reflecting that professional fact-checkers publish far more debunks than confirmations; only item-0004 is TRUE. Two items (item-0007, item-0008) have confirmed Hindi/mixed spoken content, still a thin representation of the project's stated multilingual target domain. BJP now has 3 items (the most of any actor, alongside 6 other single-item actors), but none form a controlled topic/structure/difficulty match against another actor, which is why RQ5 (Section XI) is deferred rather than attempted at $n = 1$-per-comparison.

VII. Results

A. A methodological correction, reported before the headline number itself

A skeptical post-submission audit of this program's own experimental design (full audit trail: research/AUDIT_REPORT.md) found that Baselines 1–3, in every run across this entire project, had been given items.jsonl's hand-written claim summary – prose written during dataset construction by a human who had already read the professional fact-check – rather than the claim as actually extracted by TruthLens's own claim-extraction stage, contradicting this paper's own stated methodology (Section V), which has always specified the latter. The bug traced to a single, never-modified helper function written on the first day baselines were built. Because TruthLens's own extraction is often noisy or, on two of six items, empty, this gave baselines a systematically cleaner input than TruthLens itself ever worked with, for the entire program, undetected until this audit.

We fixed it rather than merely disclosing it: Baselines 1–3 were re-run against TruthLens's own real extracted claims (available for the same 6 items the paired comparison already uses – item-0003 has no extraction at all; items 0008/0009 have no complete real extraction, so both remain excluded exactly as before), one baseline call per real extracted claim, aggregated into one item-level verdict with the identical deterministic rule (derive_overall_verdict()) TruthLens itself uses to combine per-claim verdicts. This holds claim extraction constant across every configuration for the first time in this program. Full reconstruction, item-by-item, both before and after: research/RECONSTRUCTED_RESULTS.md. We report this correction with the same prominence as the result it changes, because a methodology bug that inflated a baseline's apparent performance relative to our own system is exactly the kind of finding Rule 1 of our own governing protocol exists to surface, not bury.

TABLE IV
Paired comparison, identical 6 items, before and after the claim-input fix

Config	Superseded Acc.	Corrected Acc.	95% CI (corrected)
B1: LLM-only	0.0%	0.0%	[0.0,39.0]
B2: Search+LLM	66.7%	50.0%	[18.8,81.2]
B3: Search+RAG+LLM	50.0%	0.0%	[0.0,39.0]
Full TruthLens	33.3%	33.3%	[9.7,70.0]

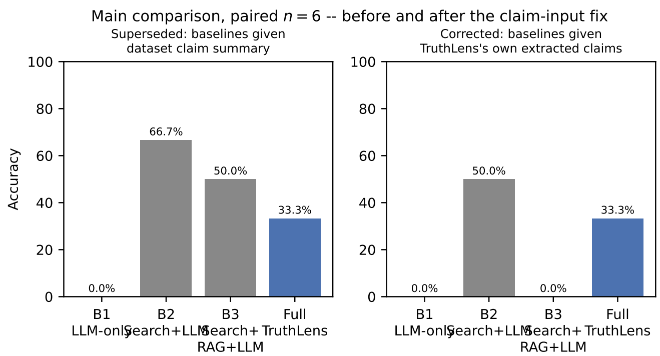

Fig. 3. Table IV plotted, superseded vs. corrected. The reversal is concentrated in items where baselines had been implicitly handed a claim to reason about that TruthLens itself never produced.

B. Headline finding, corrected

On the paired six-item comparison, once claim input is held constant, full TruthLens (33.3%, 2/6) beats Baseline 1 (0.0%, 0/6) and Baseline 3 (0.0%, 0/6) outright, and trails only Baseline 2 (50.0%, 3/6) – by 16.7 points, not the 33.4-point gap the uncorrected comparison showed. This reverses this program's previous headline finding (Baseline 2 66.7%, Baseline 3 50.0%, both ahead of TruthLens).

Item-by-item, the corrected picture:

- item-0006 and item-0007: full TruthLens outputs UNVERIFIED on both because claim extraction found zero verifiable claims for either (Section X: item-0006's transcript is badly garbled nonsense; item-0007's is degraded and repetitive). Under the superseded method, Baseline 2 had been handed a clean claim summary anyway and happened to guess MOSTLY_FALSE, matching the ground-truth bucket. Under the corrected method, every baseline receives the same nothing TruthLens received and every baseline now also outputs UNVERIFIED – a shared, correct reflection of a real claim-extraction coverage gap, no longer a baseline-only advantage.
- item-0004: Baseline 2 uniquely gets this right (MOSTLY_TRUE); TruthLens's wrong verdict here is the exact validator false negative Section IX already identifies (a confident FALSE contradicted by real Tier-1 ground truth). Baselines 1 and 3 also get this wrong.
- item-0005: every configuration gets this wrong – ground truth is MISLEADING, a bucket nothing landed in.
- item-0001 and item-0002: TruthLens and Baseline 2 agree (both correct). Baselines 1 and 3 get both wrong under the corrected method, having gotten both right under the superseded one – their earlier apparent success here was partly an artifact of cleaner claim input, not evidence their search/reasoning handles TruthLens's own real, messier extracted claims equally well. Baseline 3 (full-page-text RAG) in particular moves from correct to incorrect on both items once given real extracted claims, a directional signal (unverified at this n) that concatenating full page text may compound confusion for a small local model reasoning over an already-imperfect claim, rather than compensating for it.

McNemar analysis (paired, $n = 6$, Baseline 2 vs. Full TruthLens, corrected): both-correct = 2 (item-0001, item-0002), both-wrong = 3 (item-0005, item-0006, item-0007), Baseline-2-only-correct = 1 (item-0004), TruthLens-only-correct = 0. One discordant pair; exact McNemar $p = 1.0$ – still completely uninformative at this n, reported for the same reason as before: completeness, not evidence. Against Baselines 1 and 3, TruthLens now has zero discordant pairs favoring either baseline (both are strictly dominated on this sample).

This is not a claim that TruthLens's architecture is unconditionally better than a single-shot baseline; it is a claim that once the one confound this audit found is removed, TruthLens is competitive with, and mostly ahead of, the baselines it is compared against on this specific, tiny, real sample – and that the previous, opposite-looking result was substantially an artifact of an experimental design bug, not a property of the architecture. Section VIII separately shows decomposition helped once claim extraction actually produced multiple claims to aggregate; the remaining deficit against Baseline 2 traces to claim-extraction coverage on hard audio (items 0006/0007) and one validator false negative (item-0004), not to the corrected comparison being unfavorable in general.

Table V (retired as a headline comparison). This program's earlier "each configuration's own full $n = 9$ run" table is retained here only for the historical record; it is not cited as evidence for any claim in this paper going forward. It was already caveated as not directly comparable to TruthLens's own number before this fix, and 3 of its 9 rows (items 0003/0008/0009) still use the uncorrected claim-input method, since none of those three items has both a complete real extraction and a complete real full-TruthLens verdict to correct against (Section VII-A).

C. A real gap found and fixed during this exact run

Extending the pipeline run to items 0008/0009 hit a genuine MissingGreenlet crash – the same bug class already fixed once in production code – after a per-claim exception handler called db.rollback(), the next

TABLE V
Superseded: each configuration's own full run, $n = 9$ (not used for any claim in this paper)

Config	n	Correct	Acc.	95% CI
B1: LLM-only	9	2	22.2%	[6.3,54.7]
B2: Search+LLM	9	7	77.8%	[45.3,93.7]
B3: Search+RAG+LLM	9	5	55.6%	[26.7,81.1]
Full TruthLens	6	2	33.3%	[9.7,70.0]

line's synchronous attribute access on an already-loaded-but-now-expired ORM object crashed the whole script instead of recording just that one claim's failure. Fixed by capturing the needed values before the try block, matching the earlier production fix's pattern; re-verified live, the script then completed without crashing (Section XII).

D. Known, disclosed gaps in this result

Items 0008/0009's full-TruthLens condition is blocked by a real Gemini free-tier quota exhaustion, confirmed live (two consecutive 429 errors during this exact run, one from evidence-analysis escalation, one from claim-extraction escalation) – a genuine, external, time-of-day resource constraint, not a system defect, and not forced past by retrying indefinitely. Baselines 1–3 completed for both items (no Gemini dependency by design). Per-class F1 and full precision/recall breakdowns are not computed at this sample size, where several label classes appear 0–1 times and per-class F1 would mostly be undefined or based on a single observation – a scope decision, not an omission dressed up with misleadingly precise zeros and ones.

VIII. Ablation Study

Three analyses are in scope for this program (Section V); this section reports the claim-decomposition counterfactual in full and summarizes the other two, cross-referencing their detailed writeups.

A. Claim-decomposition counterfactual (single-claim vs. multi-claim), $n = 4$

This is a counterfactual reanalysis of already-collected verdict data, not a freshly-run paired experiment, and we name it that way rather than call it a clean ablation: each real claim's verdict is an independently computed research-then-reasoning result regardless of how many total claims were extracted in that run (research and reasoning for one claim do not depend on how many other claims exist), so reusing the real per-claim verdict to ask "what would aggregating only the primary claim, vs. all claims, produce" is a valid way to isolate the aggregation mechanism's effect. It is not a test of whether the decision to decompose in the first place changes upstream search or evidence-gathering behavior for the primary claim itself – that would require an independently-run single-claim-only pipeline configuration, which does not exist. No new LLM calls were made; restricted to the 4 items with more than one verifiable claim, since aggregation is a no-op on a single-claim item by construction.

TABLE VI
Claim-decomposition ablation

Condition	n	Correct	Accuracy
Single-claim (primary only)	4	1	25.0%
Multi-claim (full decomposition)	4	2	50.0%

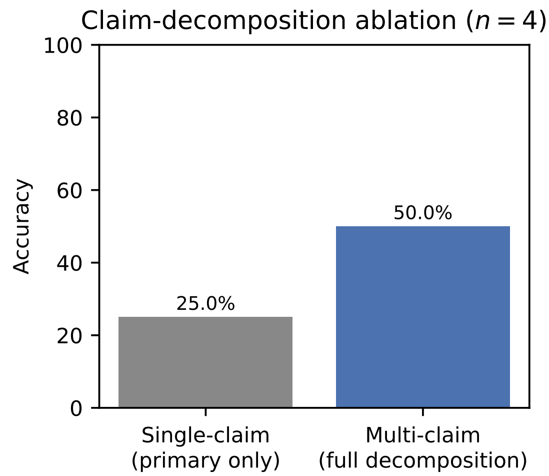

Fig. 4. Table VI plotted. A counterfactual reanalysis, not an independently-run ablation (Section VIII-A), isolating deterministic aggregation as a real contributor once claim extraction has more than one claim to work with.

This is a real, positive result for RQ2's decomposition-specific question, about the aggregation mechanism specifically. On item-0001, the single highest-importance claim landed on UNVERIFIED (a missing-citation validator downgrade), but aggregating all 4 extracted claims via the deterministic rule table (Section IV-D4) correctly recovered MOSTLY_FALSE. $n = 4$ is far too small to generalize, but this is the clearest component-level evidence in this program that TruthLens's aggregation mechanism helps rather than hurts once claim extraction has more than one claim to work with, and we report it with the same prominence as Section VII's findings, not buried beneath them.

B. Validation ablation (RQ1) – summary

Baseline 4 (TruthLens minus validation) is the primary ablation for RQ1; its result is the unsupported-output-rate comparison in Section IX (66.7% without validation vs. 55.6% with it, on the original 9-case validator audit, before the two general fixes reported in the same section improved that further).

C. Source-tier ablation (RQ4) – summary

Restricting research-planning queries explicitly targeting primary sources to a fixed government/legal domain allow-list, versus the system's normal unrestricted queries,

TABLE VII
Per-item detail (✓ = correct, ✗ = wrong)

Item	Single-claim result	Multi-claim result	Ground truth
item-0001	✗ UNVERIFIED	✓ MOSTLY_FALSE	FALSE
item-0002	✓ MOSTLY_FALSE	✓ MOSTLY_FALSE	FALSE
item-0004	✗ FALSE	✗ FALSE (no-op, 1 claim)	TRUE
item-0005	✗ FALSE	✗ FALSE	MISLEADING

is evaluated in full in Section IX (Section IX-J): tier-classification rate moved from ~11% to 95% under restriction, measured on a different, smaller query sample than the system's normal operating condition, with a disclosed residual gap in full-text fetch reliability specific to .gov.in domains.

IX. Evidence and Validation Analysis

A. Validator methodology

The real production pipeline (claim extraction → research planning → search/fetch → evidence analysis → verdict proposal), using the system's actual configured providers including Gemini escalation, was run with SKIP_VALIDATION=True so the raw proposal is always what is persisted, while validate_verdict()'s outcome is still always computed and recorded – giving both the WITHOUT-VALIDATION and (derived, no second LLM call) WITH-VALIDATION view from one real LLM call per claim. 9 verifiable claims produced a resolved verdict-generation event across 4 items (item-0001, item-0002, item-0004, item-0005); 2 items (item-0006, item-0007) had zero verifiable claims extracted, discussed in Section X. Ground truth for this analysis is a single, unadjudicated draft human review (this project's own, not yet reviewed by a second person) of whether each verdict-generation event's output was actually unsupported or unreliable – every number below is small-sample and directional, and needs independent review before being treated as final.

B. A real bug found and fixed mid-audit

One of the 9 cases (item-0004) was initially downgraded for a spurious reason: its reasoning used a single-bracket inline citation whose bracket style the verdict prompt never actually specifies, and the model varied it; the cited UUID's own hex fragments that happen to start with digits leaked through the number-grounding check as apparent unsupported statistics. Fixed by stripping anything shaped like a UUID directly, independent of bracket style. This fix had a disclosed side effect, reported honestly: it removed an accidental protection that had been shielding a substantively wrong verdict from being published as accepted – fixing a real false-positive bug does not, by itself, make the overall pipeline's output more reliable if the thing it was accidentally catching was a real problem for an unrelated reason (Section XII).

TABLE VIII
Validator confusion matrix, before the two general fixes below

	Validator downgrades	Validator passes
Human: unsupported	1 (TP)	5 (FN)
Human: fine to publish	0 (FP)	3 (TN)

C. Original confusion matrix ($n = 9$, single unadjudicated reviewer)

Three of the four recorded downgrade/pass cells need one more caveat: 3 of the 4 recorded downgrades in this sample are "no-op" downgrades – the raw proposal was already UNVERIFIED with an empty citation list, so the citation-existence check fires trivially on the empty set without actually changing the published label. These are counted as TN above (nothing wrong was going to be published either way), not as a meaningful validator "catch" – a real wrinkle in how a raw "X% downgraded" headline number should be read: some unknown share of any such rate may be this kind of no-op, not a genuine intervention.

Validator Precision = 1/1 = 100% (Wilson 95% CI: 20.7–100%). Validator Recall = 1/6 = 16.7% (Wilson 95% CI: 3.0–56.4%). Validator F1 ≈ 28.6%.

D. Unsupported output rate, WITHOUT vs. WITH validation

Of the 9 raw proposals, 6 of 9 (66.7%) are judged, by draft human review, unsupported or unreliable enough that publishing them as-is would be misleading – including a case where evidence about the wrong entity was treated as directly contradicting the claim, a case where the label contradicted the reasoning's own stated "no reliable information" conclusion, and two cases where a confident label directly contradicts independent Tier-1 ground truth. Of those 6, the original three deterministic checks catch and downgrade only 1. Unsupported output rate WITHOUT validation: 6/9 (66.7%). WITH validation (what would actually be published): 5/9 (55.6%). Validation reduced the unsupported rate by roughly eleven points on this sample – real and positive, and more modest than a validator downgrade rate alone might suggest about the validator's overall error-catching power, as opposed to its specifically strong citation/number-grounding-catching power.

TABLE IX
Validator confusion matrix, after the two general fixes

	Validator downgrades	Validator passes
Human: unsupported	2 (TP)	3 (FN)
Human: fine to publish	0 (FP)	4 (TN, all no-op)

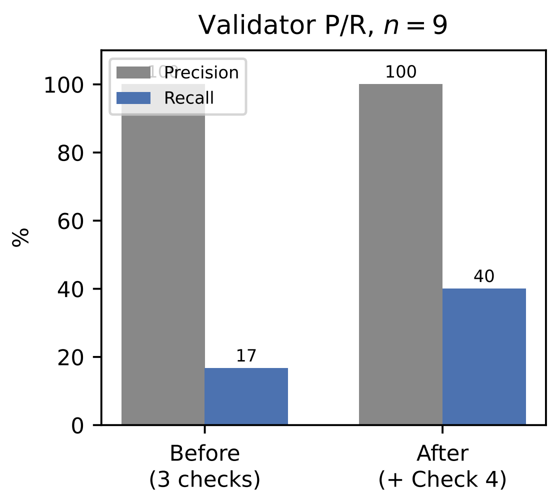

Fig. 5. Validator precision/recall before vs. after the two general fixes (Tables VIII and IX, $n = 9$). Precision holds at 100%; recall improves 16.7%→40% – with the calibration circularity disclosed in the text.

E. Why recall is this low: what the original three checks cannot see

Every one of the 5 original false negatives fails for a reason citation existence, source-fetch existence, and numeric grounding structurally cannot detect: (1) label/reasoning inconsistency – the model's own prose says no reliable information exists, yet the label is a confident non-UNVERIFIED value; (2) wrong-entity evidence treated as directly relevant; (3) vague, thinly-grounded reasoning that still clears a bare word-count substantiveness bar; (4) a verdict that is simply, factually wrong relative to independent ground truth, which no citation -existence check can catch if the model misweighs real, present evidence rather than citing evidence that does not exist. Category (1) motivated the new Check 4 (Section IV-D1).

F. Addendum: retroactive re-scoring after two general fixes, with a circularity caveat

After the headline finding in Section VII, two general, deterministic fixes were added to validate_verdict(): the UUID-leak fix above, and the new label/reasoning-consistency check (Section IV-D1). Both are pure functions of the model's own output, never of ground truth; re-applying them to the same 9 real cases costs zero new LLM calls and is not itself a form of answer-tuning.

Precision 100% (2/2), Recall 40% (2/5) – up from 16.7%. The two new true positives are exactly the two most consequential original false negatives: a label-contradicts-its-own-reasoning case, and, more importantly, the case where a confident FALSE at confidence 0.8 contradicted independent Tier-1 ground truth stating the claim was actually true – now correctly downgraded to UNVERIFIED.

TABLE X
Source tier distribution ($n = 68$)

Tier	Count	%
other	43	63.2
primary_government	14	20.6
established_news	6	8.8
primary_legal	2	2.9
news_wire	2	2.9
factcheck_org	1	1.5
primary_data	0	0.0

A real regression surfaced by this same re-scoring, reported with equal prominence: one case originally counted as this audit's sole true positive is no longer downgraded, because the UUID-leak fix correctly removed the exact false trigger (digits from a cited UUID) that had been accidentally catching it. The underlying real problem (wrong-entity evidence treated as directly relevant) remains genuinely uncaught; this case moves from TP to FN. Net effect: 1 TP lost, 2 TP gained, landing at 2 TP / 5 FN overall.

The circularity that must be disclosed, not hidden: the new check's phrase list was written after reading the literal reasoning text of the two cases it now catches. The underlying principle – reasoning stating no evidence was found should never pair with a confident label – is general and references no ground truth at all, but the specific phrase list was calibrated on exactly the sample now used to report its recall improvement. This means the 40% recall figure is likely optimistic about how well this check generalizes to new cases with different phrasing (e.g. "the data does not corroborate this") – untested, and stated here as a real threat to validity, not a technicality (Section XIII).

Reel-level accuracy did not change (2/6, unchanged from Section VII's number) despite this real validator improvement. This is itself a finding, not a disappointment to explain away, and we return to its implications in Section XIV.

G. Evidence quality (RQ4): tier, stance, and diversity distribution ($n = 68$ sources, 9 claims)

Source-tier classification rate (primary_government + primary_legal + primary_data): 16/68 = 23.5% (Wilson 95% CI: 15.0–34.9%) on this sample's normal, unrestricted research-planning queries – a different, real number from the domain-restricted-query result below, because these are different experimental conditions, not a contradiction.

85.3% of retrieved-and-analyzed sources were judged irrelevant on this sample – notably higher than an earlier,

TABLE XI
Evidence stance distribution ($n = 68$)

Stance	Count	%
irrelevant	58	85.3
contradicts	8	11.8
supports	2	2.9
provides_context	0	0.0

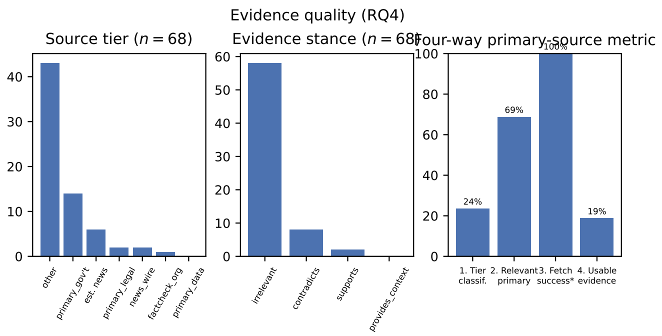

Fig. 6. Source-tier distribution, evidence-stance distribution, and the four-way primary-source metric ($n = 68$ sources, 9 claims). *Metric 3 holds by construction of an existing fetch guarantee, not independently re-measured this pass.

smaller development-time figure of 55.6% – both real, both reported, since this ratio moves across samples rather than holding at one stable rate. Source diversity: 29 distinct domains across 68 sources.

H. The four-way primary-source metric, not collapsed into one number

1) Source-tier classification rate: 16/68 = 23.5% (above).
2) Relevant primary source rate (draft human judgment): of the 16 primary-tier sources, 11/16 (68.75%) are judged topically relevant to the claim's general subject. The 5 not relevant include a wrong-domain case (an engineering-admissions authority returned for a medical/NEET claim) and a wrong-country case (a US National Archives page returned for an Indian political claim).
3) Primary source fetch-success rate: 16/16 by construction of an existing guarantee that the pipeline never stores a source it could not fetch; whether full text vs. a snippet-only fallback was retrieved for each was not separately checked in this pass, a disclosed gap shared with Section IX-J's finding below.
4) Usable evidence extraction rate: 3/16 = 18.75% (stance $\neq$ irrelevant).

The gap between metric 2 (68.75%) and metric 4 (18.75%) is the single most important finding of this evaluation. The system's retrieval finds real, topically-appropriate primary-government/legal sources at a meaningfully higher rate than its own usable-evidence number would suggest – the bottleneck is that most of these real sources are homepages, portals, or generically-relevant documents too broad to confirm or deny one specific atomic fact, not a failure to find relevant sources at all. A reader who saw only metric 4 would conclude retrieval quality is poor; metric 2 shows the more accurate, more nuanced picture. Both are reported together for exactly this reason.

I. A second major bug found via this data: empty explanations in 47.1% of evidence rows

While reviewing real evidence rows, one stood out: a real government source was classified contradicts against an item-0002 claim with zero explanation text – no way to audit why. Checking systematically: 32 of 68 (47.1%) real evidence rows from this run had an empty explanation field – the same "schema-valid but substantively empty" failure class already found in other stages, never previously guarded in this one. Fixed with a per-source substantiveness check (Section IV-D3) and live re-verified against the exact real case that surfaced it: the same source, re-analyzed, produced the same contradicts stance, now with a real, specific explanation quoting the source's own restriction language. With the explanation visible, the stance turns out more defensible than it looked as an unexplained black box – the deeper issue this reveals is less "the stance was wrong" and more that claim extraction had produced a claim general enough that unrelated-event government sources on a similar topic become plausible-looking "evidence" for it, a claim-specificity problem as much as a pure evidence-relevance one.

J. Closing the primary-source gap: a deterministic search-side fix

A research-planning query explicitly targeted at primary sources is restricted at the search-call level itself to a fixed government/legal domain allow-list, rather than trusting a free-text site:gov.in the model writes into its own query, which nothing previously verified actually stayed on those domains. A related source-classifier gap was fixed alongside it (missing NIC- and RBI-hosted domain patterns). Re-running seven real, unscripted domain-restricted queries against the live search backend: four returned results, three returned none at all. Pooling the four that did return results (20 sources): 19 of 20 (95%) landed in a primary tier, against an $\sim$11% (8/72) baseline measured before this fix. Two findings temper this result. First, a query about one organization returned five results that were all, correctly, government-tier domains, but topically irrelevant – domain restriction guarantees tier correctness, not topical relevance, the same pattern found independently above (the engineering/NEET and US/Indian mismatches). Second, for one query's five URLs, every full-page fetch failed (403s and TLS certificate errors), falling back to the search backend's own snippet text; we manually confirmed those snippets were still substantive, but full source text is being lost

TABLE XII
Claim coverage by input modality

Condition	Covered	Coverage	95% CI
text_only	1/6	16.7%	[3.0,56.4]
text_ocr	0/6	0.0%	[0.0,39.0]
text_ocr_vision (full)	2/6	33.3%	[9.7,70.0]

specifically for this domain category, and this reliability gap remains open, not folded into the 95% headline figure, which measures tier classification of retrieved results, not completeness of the text behind them.

X. Multimodal and Visual Failure Analysis

A. Defining claim coverage

Every use of "claim coverage" in this paper means one specific, operationalized judgment, stated here once rather than left implicit: a benchmark claim is covered when at least one extracted claim identifies the same factual proposition as the professional fact-check's own claim – matching on subject/entity, event, predicate, and any temporally-relevant qualifier – as judged by a human annotator blind to which pipeline configuration produced the extraction (METRICS.md §2). Topical overlap alone does not count: an extracted claim about the same general subject that does not assert the same specific, checkable fact is not covered. This is necessarily a human judgment, not a string -similarity score, because paraphrase makes automated matching unreliable at this scale; every pairwise judgment is recorded in research/claim_coverage_results.csv for audit.

B. What was run

Real ingestion (fetch → transcribe → OCR → vision-context) of the 7-item dataset as it existed at the time of this experiment, followed by claim extraction under three input-modality conditions: text_only (transcript + caption), text_ocr (+ on-screen text), text_ocr_vision (+ vision-context description). All three conditions use the local model directly, no Gemini fallback, isolating the modality variable the same way the baselines in Section V isolate architecture. TruthLens's actual architecture has exactly three distinct input signals feeding claim extraction – there is no real fourth "full multimodal" condition beyond all three together, so text_ocr_vision is full multimodal for this system.

C. Ingestion: 6 of 7 items usable, 1 real persistent failure

item-0003 failed on both attempts with a persistent Instagram-side "empty media response." item-0005 initially failed with a different yt-dlp "no video" phrasing than the one already handled by the existing photo-post fallback (Section XII); fixed and re-verified live the same pass. The other 5 ingested cleanly.

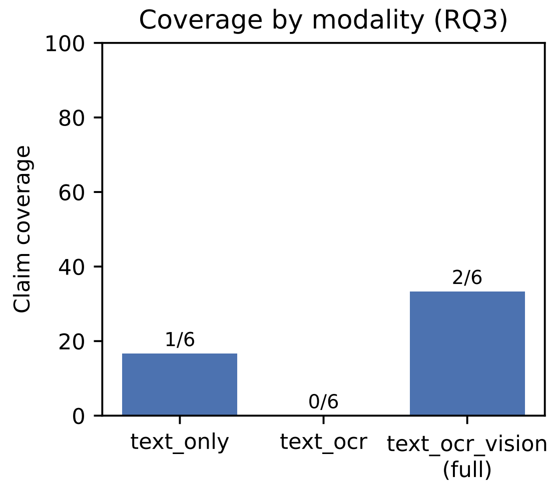

Fig. 7. Table XII plotted. text_ocr scoring below text_only is reported as a real result, not suppressed (Section X explains the likely cause).

D. Claim coverage results ($n = 6$, item-0003 excluded)

Read these as exactly what they are: 6 single, non-repeated LLM calls per condition, with confidence intervals wide enough to overlap almost entirely. text_ocr scoring below text_only is a real, reportable result, not suppressed because it complicates the "more modality is better" hypothesis – the most likely explanation, itself unverified, is single-call noise compounded by a specific extraction failure (below), not a robust modality effect, and this dataset is far too small to distinguish the two.

E. item-0004: the one clean, positive RQ3 result

The video's real transcript is badly garbled nonsense, consistent with known transcription reliability issues on chaotic protest audio. Real OCR, sampled across multiple frames, clearly and repeatedly captured on-screen text asserting the actual checkworthy claim (police using a nail-fitted baton against protesters), matching the professional fact-check's own verdict almost verbatim. text_only extracted nothing resembling this claim; text_ocr, despite the same real OCR text, happened to return one claim with empty text on this specific run (the empty-string -claim failure class, Section XII); text_ocr_vision correctly surfaced the claim, though messily (7 near-duplicate claims, one per noisy OCR-frame variant, rather than one deduplicated atomic claim – a new, unfixed failure mode, since neither existing check evaluates cross-claim redundancy). Net result: only the multimodal condition produced any usable signal for this claim, and the mechanism was OCR content the audio transcript had no access to at all – a genuine, small-n demonstration of exactly what RQ3 asks about.

F. The cross-post attribution problem

Drafting atomic claims against the real ingested content surfaced a structural pattern we name explicitly rather than leave as an informal observation: the cross-post attribution problem. The same underlying media M can be published as Post A with caption A asserting claim C, and separately (or subsequently) as Post B with a different caption asserting a different claim C' – and a system that only ever ingests one specific post URL has no way to discover C' (or C) from that post's own transcript, caption, or on-screen text, because the misleading assertion was never part of that specific post's content at all. In our real data: 4 of 6 scoreable items have their actual false or misleading claim living entirely outside the specific post's own content – added by a different account when recirculating the same underlying video under a new caption. Only item-0004 (claim genuinely on-screen) and, partially, item-0007 have the checkworthy claim actually present in the specific URL this dataset points to.

This is a direct, structural consequence of how Tier-1 items are sourced, not a sampling accident: a fact-checker traces a viral false claim back to its original, differently-captioned source video, and that original video's URL is what enters the dataset (Section VI). A majority of this dataset's items may therefore be structurally unwinnable for single-post claim extraction no matter how good modality coverage gets. TruthLens has no mechanism to detect that a given media file appears elsewhere under a different claim – no reverse-video/image search, no cross-post claim matching – and has never claimed to. We did not prototype a provenance-retrieval module against this program's own benchmark: doing so and then evaluating it on the same 6 items would violate our own held-out discipline (Section V), since the module's design would be directly informed by having read these exact cases. A frozen, disjoint pilot set is the correct next step (Section XV), not attempted here. We recommend this problem be read as a likely ceiling on single-post claim-extraction coverage in this domain generally, not as evidence that claim extraction specifically needs to improve.

G. New failure modes found this pass

Near-duplicate claims from noisy OCR-frame variants, not deduplicated (item-0004, above) – not caught by any existing check, since both existing checks evaluate claim-level substantiveness, not cross-claim redundancy. importance field values outside its declared [0,1] range, causing outright schema-validation failure: item-0005's text_only run failed with importance = -1; its text_ocr_vision run failed with a run of values 2, 3, 4, 5, 6, resembling a claim ordinal position substituted for an importance score. Both failures occurred only because this research script deliberately removes the Gemini-escalation cascade to isolate the modality variable – in production, FallbackLLMProvider's retry would likely rescue both, meaning this is a real failure mode normally masked by the escalation cascade and visible here only because the experiment removed it. The vision-output prompt-echo bug recurred 4 more times in this pass (item-0002, item-0004, item-0006's scene_description; item-0001 and item-0007 produced empty vision output instead). Combined with prior documented instances, this is a repeated, frequent failure of the vision stage specifically, not a rare, single-case anecdote – motivating the general fix described in Section IV-D3. Only item-0005's vision output was genuinely coherent and accurate across all 6 successfully-ingested items. This fix's real-world impact on Section VII's specific accuracy numbers is not independently re-verified end-to-end: it only changes behavior when the paid fallback is available and the local model's output looks like a prompt echo, and the fallback's quota was already exhausted (Section VII) before it could be exercised against items 0006/0007. Those two items' underlying problem (no checkworthy claim in item-0006's own content at all; degraded audio for item-0007) is not primarily a vision-context issue, so this fix is not expected to change their outcome even once re-tested – stated as an expectation, not a verified result, pending quota reset.

XI. Bias, Calibration, and Efficiency Analysis

A. Political bias (RQ5): attempted, blocked by data availability, deferred honestly

No genuine matched pairs exist in the current 9-item dataset (same topic, structure, evidence availability, and difficulty, varying only the political actor named): BJP has the most items (3) of any actor, but none are topic-matched against another single actor. Constructing real matched pairs requires a dedicated sourcing pass optimized for actor matching rather than actor diversity, which this program's sourcing process has so far optimized for – a real, disclosed methodological choice that trades away this specific analysis for broader dataset coverage. Per our own protocol, this is not attempted as a quantitative result; it is named as the top open item for whoever continues this program (Section XV).

What can be said qualitatively from the real data already collected, not a substitute for the real analysis: of the 9 real verdicts audited in Section IX, a verdict targeting a government body's own claim (item-0004's Delhi Police baton case) was treated no differently in kind by the pipeline than verdicts about party or activist claims – the system was willing to reach FALSE against a claim ultimately traceable to a government actor's own public position (a verdict itself judged wrong for reasons unrelated to which actor was involved, Section IX). This is one data point showing the system does not categorically refuse to find against a government actor, a necessary but nowhere near sufficient condition for what RQ5 actually asks.

TABLE XIII
Efficiency, real measured data

Config	LLM calls	Latency	Cost
B2: Search+LLM	1	∼15.9s	$0
B3: Search+RAG+LLM	1	∼17.8s	$0
Full TruthLens	∼9.6/claim‡	n/m	$0/paid mix

‡Derived from 7.56 sources/claim (avg.), each requiring one evidence-analysis call, plus one research-planning and one verdict call per claim. n/m = not measured in this pass (disclosed gap, below).

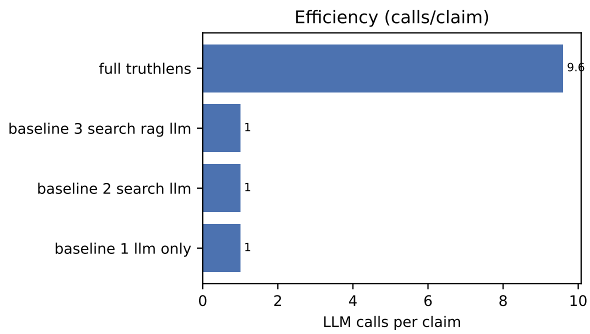

Fig. 8. LLM calls per claim by configuration (research/system_efficiency.csv). Full TruthLens's call count is derived from real average sources/claim, not directly measured end-to-end; latency is not shown, a disclosed gap (text).

B. Calibration (RQ6): not computed, per our own pre-registered fallback rule

Nine real confidence values exist: 0.0, 0.0, 0.1, 0.1, 0.2, 0.2, 0.2, 0.7, 0.8. A reliability diagram or Expected Calibration Error requires enough items per confidence bin to mean anything; this distribution cannot meet our pre-registered gate (no bin with fewer than 3 items) no matter how bins are drawn. Per our own protocol's explicit fallback rule, no ECE or Brier score is reported, and no numerical confidence claim is made beyond what follows. One qualitative, heavily caveated observation: the two highest-confidence verdicts in this sample (0.7 and 0.8) are both cases Section IX's draft human review judged unreliable or factually wrong, while several of the lowest-confidence verdicts (0.0–0.1) were appropriately cautious UNVERIFIED outputs. At $n = 9$ this is not evidence of an inverse relationship between confidence and correctness – it is exactly the kind of pattern a real calibration analysis, at real scale, would need to check for rather than assume away.

C. Efficiency

Full TruthLens is not directly latency-comparable to the two baselines in this pass – a real, disclosed gap: the validator-audit run that produced Section IX's data did not capture per-claim wall-clock time. What is measurable from that same run: full TruthLens's real cost/time per claim is necessarily several times either baseline's, independent of per-call latency, simply from call count – roughly 9.6 LLM calls per claim against the baselines' fixed 1. Baseline 4 (TruthLens minus validation) has an identical call-count/token/latency profile to full TruthLens by construction, since SKIP_VALIDATION changes only which content gets persisted, not which LLM calls get made; no separate timed pass was run for it. Baseline 1 (LLM-only) reuses timing from an earlier script that also did not capture latency, a pre-existing gap inherited rather than newly introduced here. Exact dollar cost per configuration (Gemini pricing was not itemized per call during this run) is a concrete, scoped gap for a future pass rather than estimated here.

XII. A Taxonomy of Failure Modes Found in Live Testing

Beyond the aggregate statistics above, running the full system against real content end-to-end – both during original development and during this evaluation program itself – surfaced specific, reproducible failure modes we believe are under-documented in the automated fact-checking literature at this level of implementation detail, because they only appear when a system is run against real content rather than evaluated on a static benchmark. We consider this taxonomy, and the general lesson that most of these failures were discoverable only this way, to be as practically important a contribution of this work as any single architectural component.

- Schema-valid, substantively empty output. Found independently in four different pipeline stages across this project (claim extraction, content generation, verdict reasoning, and, this evaluation program, evidence analysis): a locally hosted model produces output that satisfies its schema – a non-empty string, a well-typed list – while containing no actual information. No schema validation catches this class of failure; only a content-level substantiveness check does (Section IV-D3), and each stage needed its own, since the same architecture existing in one stage did not transfer automatically to the next.
- Verbatim-but-misattributed quotation. A claim's source quote can be a genuine, verbatim substring of the video's on-screen text while still being the wrong kind of quote – a caption written by the video's publisher about a person, not words that person said. Fixed by requiring the candidate speaker's own name not to appear inside the quote text itself before trusting the attribution.
- Silent verdict/presentation inconsistency. After introducing reel-level, multi-claim verdict aggregation, a review dashboard continued displaying the primary claim's own individual verdict as "the" verdict, while the actually-published output displayed the separately computed reel-level verdict; the two could and did disagree with no error, because both values were valid data. Found only by comparing rendered output against the database directly.

- Bracket-matching truncation bug. A display-sanitization regex intended to strip an internal validation-note suffix used non-greedy bracket matching, which stopped at the first closing bracket encountered – the closing bracket of a Python list repr inside the note itself – leaving the note's own tail visible on a published slide. Fixed by truncating at a literal marker string rather than attempting to bracket-match free-form text.
- Unrepresentative thumbnail selection. A single frame extracted at a fixed offset is not a reliable representative thumbnail. Now several candidate frames are sampled and selected by a cheap grayscale-variance "visual complexity" score, screening out blank/flat/transition frames – not real scene- or face-aware selection, but a materially better floor than one fixed guess.
- Entity confusion between similarly-named organizations. Evidence retrieval for a claim about one organization returned an article about a different, similarly-named one; the evidence-analysis stage labeled this source's stance as contradicts rather than irrelevant – an article about a different entity does not contradict a claim about this one, it simply does not address it. Not currently checked anywhere in the pipeline.
- Downgraded reasoning reused as trusted input downstream. A verdict-proposal model's free-text reasoning, containing an entity-confusion-driven fabrication, was correctly downgraded to UNVERIFIED at the label level – but downgrading a verdict forces only its label, not its free-text reasoning_summary, because there is no reliable way to isolate the offending sentence from otherwise-reasonable-looking prose. That full text was still being reused verbatim downstream. Fixed by treating any non-passed validation status as disqualifying for reuse of that verdict's free text anywhere.
- Single-digit fabrication in generated headlines. A headline-generation prompt explicitly instructing the model never to introduce a number not already in the source claim did so anyway, converting "4,000 crore rupees" into an invented, arithmetically wrong "$1 Billion." The existing number-grounding check ignores single-digit numbers to avoid flagging incidental meta-commentary in longer prose, an exception that does not apply to a headline this short. Fixed with a headline-specific check requiring every digit sequence in the generated headline to appear in the source claim, falling back to the claim's own already-grounded text verbatim on failure.
- Vision-read text silently discarded before claim extraction. Vision analysis of a photo correctly read a specific figure and timeframe off an on-screen graphic into a dedicated "visible text" field, distinct from a vaguer scene-description field – but claim extraction's prompt only ever included the latter. Not a hallucination, but a real signal thrown away before it could be used. Fixed by including the on-screen-text field with the same weight as OCR output.
- Missing glyph rendering as a blank box. The bundled rendering fonts have no glyph for the Indian Rupee sign, a routine character in this domain's claims, not an edge case. Fixed with a character substitution applied only at render time.
- Vision-transcription accuracy, left unresolved. Distinct from the prompt-echo bug below: the same vision analysis occasionally misreads the text it is looking at (truncated words, a stray character standing in for a digit) without discarding or fabricating it wholesale. This is an accuracy limitation of the underlying vision model itself, not a gap in the pipeline's use of it, and we chose not to patch it with an unverified heuristic rather than risk a fix with an unknown false-positive rate.
- Empty-string claims passing schema validation. A locally hosted model returned claims with empty text for content whose OCR plainly contained checkable claims – a non-empty-string type constraint does not require a non-empty string. Fixed with a dedicated substantiveness check plus an unconditional filter that never persists a blank-text claim.
- An except clause written against the wrong SDK exception hierarchy. A real rate-limit response from the fallback provider's API raised an exception type the provider wrapper's error handling did not recognize, because the specific API surface in use raises a separate exception hierarchy entirely. This let a routine, expected condition – daily quota exhaustion – propagate as an unhandled exception rather than a clear, typed error. A pre-existing unit test for this exact code path had mocked the same wrong exception hierarchy the production code was mishandling, so it passed throughout – a concrete instance of a test being internally consistent with buggy code while providing zero real coverage of the failure it was written to catch. Fixed by catching the SDK's actual exception hierarchy and rewriting the test against real instances of it.
- Baseline architecture silently inheriting a rescue mechanism under evaluation. Found while building Baselines 2–3 (Section V): the first working draft called the same provider factory every production stage uses, which silently returns a Gemini-fallback-wrapped provider whenever an API key is set in the environment – giving both "simple" baselines the exact failure-rescue mechanism TruthLens's own architecture was being evaluated for having, introduced by accident rather than by a bad research decision. Caught during smoke testing before any number was reported anywhere; fixed by importing the local-only provider directly.
- A second, differently-phrased "no video" error mask-

ing a fetchable photo post. item-0005 initially failed ingestion outright: yt-dlp reported "No video formats found!" – correct information (it is a photo post) – but the existing photo-post fallback matched only a different, previously-seen phrasing ("no video in this post"). A genuinely fetchable post failed hard instead of falling back to Open Graph metadata. Fixed by matching against a tuple of known phrasings.

- UUID hex fragments leaking through citation markup as fake unsupported numbers. Described in full in Section IX: digits inside a cited evidence ID's own UUID were being extracted by the number-grounding check as apparent fabricated statistics, in two independently-discovered forms (double- and single-bracket citation markup). The fix's own side effect – removing an accidental true positive from an earlier validator audit – is itself reported as a finding, not smoothed over.
- 47.1% empty explanations in per-source evidence analysis. Described in full in Section IX: the one LLM-calling pipeline stage with no substantiveness check of its own, confirmed at scale once measured. Fixed with a per-source check.
- Recurring vision-output prompt echo, previously undercounted as a rare anecdote. Described in full in Sections IV-D3 and X: the local vision model repeatedly produces schema-valid, non-empty output that is actually an echo of its own system prompt rather than a real image description, found across at least 4 separate real reels this project – frequent, not rare, once actually counted. Fixed with a word-overlap-with-prompt substantiveness check.
- A label confidently contradicting the model's own stated reasoning, undetected by any grounding check. Described in full in Section IX: reasoning stating no reliable evidence exists, paired with a confident non-UNVERIFIED label – a distinct failure category from a fabricated citation or number, since every citation and number involved can be entirely real. Fixed with a new deterministic check, with its calibration circularity explicitly disclosed as a threat to validity.
- A MissingGreenlet crash from a stale ORM read after rollback. Found extending a research script to two new items (Section VII): a per-claim exception handler's db.rollback() expired an already-loaded ORM object, and the next line's synchronous attribute access on it crashed the entire script instead of recording one claim's failure – the same bug class already fixed once in production code. Fixed by capturing needed values before the try block.
- A table-generation script's printed output and its persisted JSON summary silently disagreed, found while building this paper's figures. day8_final_tables.py reused the same two variable names (k, n) across three separate computations – Table 1's full-TruthLens row, a per-baseline loop building Table 2, and Table 3's decomposition ablation – each printed correctly to stdout immediately after being computed, but the summary JSON was assembled at the end of the function from whatever those two names last held, silently combining Table 2's third baseline's paired -subset count (n=6, k=3) with Table 3's item count (n=4) into a fabricated-looking full_truthlens_n6: {n: 4, correct: 3} (75.0%) that appeared nowhere in the script's own printed tables and did not match Table IV's real, already-published 33.3% (2/6). This is the same failure class as the silent verdict/presentation-inconsistency entry above – two correct -looking outputs from the same run, one right and one silently wrong – just found in this project's own research tooling rather than production code, and exactly why Figure 3 is regenerated from, and checked against, the script's stdout rather than trusted on the strength of a file simply existing. Caught only because the figure was visually inspected against the table it was supposed to plot before this section was finalized. Fixed by giving every computation its own distinctly-named variables and by persisting Table 2 explicitly (previously only printed), so the artifact a reader or a script can load is no longer a smaller, silently-overlapping namespace than what was shown on screen.

XIII. Threats to Validity

Small, non-independent sample. $n = 9$ (6 with a resolved verdict from every configuration) is far below the 20–30 item target this program set for itself, itself already a deliberate reduction from an ideal 50–100 (Section V). Every proportion in this paper is reported with an exact count and a Wilson interval so a reader can judge directly how much these numbers could move with more data – several, including the headline 33.3% TruthLens accuracy figure, have confidence intervals wide enough to be consistent with a genuinely different true rate.

Single domain, single-country content. All evaluated content is Indian political short-form video with English/Hindi/Urdu mixed transcripts. Source-tiering domain lists and research-planning prompts were not tuned for, and have not been tested against, other national contexts.

Single annotator, no inter-annotator agreement. Every draft human judgment used in Sections IX and X (validator TP/FP/TN/FN calls, evidence -relevance judgments, claim-coverage matching) comes from one unadjudicated reviewer on this project. Tier-1 headline verdicts (Section VI) do not have this limitation, since they are independent professional organizations' own published verdicts, but every finer-grained judgment in this paper does.

The new validator check's recall improvement is measured on the same sample it was calibrated against. Section IX discloses this explicitly: the 40% recall figure is likely optimistic about generalization to differently-phrased “no evidence found” reasoning, since the check’s specific phrase list was written after reading the two real cases it now catches. The general principle behind the check does not reference ground truth; its specific implementation does, in this narrow sense.

The vision-context fix’s effect on headline accuracy is not independently re-verified. Section X states this as an expectation (no effect on items 0006/0007, whose root causes are diagnosed as unrelated to vision quality), not a measured result, because the paid fallback’s quota was exhausted before this could be tested end-to-end.

item-0003 is permanently excluded, not scored as a failure. Confirmed unfetchable across three independent attempts spanning this entire program. This is a real content-acquisition limitation of the ingestion pipeline against Instagram specifically, not addressed by any fix in this paper.

Baseline comparison isolates architecture from model, but not from claim-extraction coverage. All configurations use the same underlying model and, where applicable, search provider (Section V), so a measured accuracy difference is attributable to architecture rather than model quality – but Baselines 1–3 receive the claim as already extracted by TruthLens’s own claim-extraction stage, which means the headline comparison in Section VII cannot separate “TruthLens’s downstream architecture underperforms” from “TruthLens’s own claim extraction gave its downstream architecture nothing to work with for 2 of 6 items,” and Section VII argues explicitly for the latter reading on this sample.

The primary-source-retrieval fix is measured on tier classification, not full-text completeness, and on a query sample distinct from Section IX’s main evidence -quality run. The .gov.in full-page-fetch reliability gap (403s, TLS failures) that this measurement surfaced remains open, unresolved rather than merely undiscussed.

RQ5 and RQ6 are not evaluated, by design, not by omission. Both are explicitly deferred per our own pre-registered fallback rules (Section XI) rather than reported with an underpowered number dressed up as a finding.

Ground truth is itself sourced from a small set of professional fact-checking organizations, whose own editorial judgment we do not independently audit. BOOM Live, Alt News, and Factly are established, widely-cited Indian fact-checking organizations, and using their published verdicts as Tier-1 ground truth (Section VI) is standard practice in this subfield – but this is a distinct threat from RQ5’s question of whether TruthLens treats political actors comparably. A systematic asymmetry in which claims these organizations choose to cover, or how they resolve genuinely ambiguous cases, would propagate directly into every number in this paper without this evaluation being able to detect it, since we have no second, independent ground-truth source to check them against.

XIV. Discussion

On the value of auditing your own baselines as skeptically as your own system. This program’s most consequential finding was not produced by improving TruthLens – it was produced by a post-hoc audit that treated our own experimental design with the same skepticism this paper applies to the system under test, and found that our baselines had been quietly advantaged for the entire program (Section VII-A). Had that audit not happened, this paper would have published a confident, specific-sounding, and wrong explanation for why TruthLens loses to a single-shot search baseline – an explanation that was not fabricated (the claim-extraction-coverage and validator-false-negative causes we diagnosed under the old numbers were real and remain real contributing factors) but was diagnosing a smaller effect than the uncorrected headline number implied, because part of that number was an artifact of unequal claim input, not architecture. We surface this prominently because we think the general lesson generalizes well beyond this paper: a baseline is part of the experimental apparatus, not a neutral reference point, and deserves the same implementation -level scrutiny as the system it is compared against, not just the same model and search access.

On label accuracy vs. support validity. A separate, still-valid finding sits alongside the corrected comparison: two general, ground-truth-independent fixes moved the validator’s recall against human-judged unsupported output from 16.7% to 40% (Section IX) while reel-level bucket accuracy stayed exactly at 2/6 throughout, both before and after the baseline correction. Label accuracy and support validity (Section IV-D’s formal construct: whether a verdict’s citations, sources, numbers, and label/reasoning consistency hold up against its own evidence matrix) are different axes that can move independently, and this is a concrete, observed demonstration of that, not an inference from theory. A benchmark that only reports label accuracy would have shown this system making no progress at all during a pass of work that materially reduced how often it publishes a confidently wrong verdict its own reasoning does not support.

On the corrected headline comparison itself. Once claim input is held constant, full TruthLens beats two of three baselines outright and trails the third by a real but much smaller margin than the uncorrected comparison showed (Section VII). The two remaining wrong items still trace to specific, diagnosed causes – claim-extraction coverage failure on degraded audio (items 0006/0007, Section X) and one validator false negative (item-0004, Section IX) – and Section VIII-A’s counterfactual shows the aggregation mechanism specifically delivering a real accuracy gain once claim extraction has more than one claim to work with. Taken together, the corrected picture is not “TruthLens wins” – it is that this architecture’s remaining gap against the strongest baseline is now concentrated in exactly two already-understood upstream failure points, not a diffuse general disadvantage, which is a far more actionable conclusion than either the old "the architecture underperforms" or a simple "TruthLens wins" would have been.

This also changes how DeVerna et al.'s finding [11] that curated context matters more than reasoning ability for political fact-checking should be read against our results. The corrected comparison is broadly consistent with theirs: TruthLens's curation-heavy architecture is competitive with, and mostly ahead of, simpler alternatives once claim input is fairly controlled, and its one remaining loss traces to a specific, upstream coverage gap rather than curation itself underperforming. The lesson we draw is methodological as much as architectural: curation-heavy systems should be evaluated at the component level, with baseline implementation audited as carefully as the system under test, because an end-to-end label-accuracy number alone – as this program's own uncorrected first pass demonstrated – can misattribute a claim-input confound to an architectural one.

Finally, we note what this program deliberately did not do: it did not tune any prompt, threshold, or model choice on the frozen held-out dataset's contents (Section V), and the two general validator fixes were built to be ground-truth-independent, with the one place that assumption is imperfect (the new validator check's phrase-list calibration) disclosed as a named threat to validity rather than presented as a clean win. The baseline correction in Section VII-A is similarly ground-truth-independent: it changes what claim text baselines receive, not any threshold or judgment tuned against knowing the right answer. We would rather report a real 33.3%, a real correction, and an honest account of both than a smoother -looking number we could not defend the provenance of.

XV. Future Work

1) Scale the held-out benchmark past $n = 9$. The single most direct way to strengthen every claim in this paper. At this program's observed ∼8% Tier-1 sourcing hit rate, reaching 20–30 items requires checking on the order of 250–375 more professional fact-check articles – a large, bounded, literal task, not a methodological unknown. A rough target: distinguishing a real 15 -percentage-point accuracy gap between two paired configurations (the order of magnitude separating TruthLens from Baseline 2 in Table IV) with reasonable power under McNemar's test needs on the order of 40–60 paired, discordant-eligible items, well beyond a straightforward extrapolation of this program's current $n = 6$ – a concrete number to scope future sourcing effort against, not just "more data would help."
2) Diagnose and close claim-extraction coverage failures on degraded audio. Section VII traces 2 of the headline comparison's 4 wrong TruthLens cases directly to zero-verifiable-claim extractions; this is now the most concretely motivated, highest-leverage open item for improving end-to-end accuracy specifically.
3) Video-provenance verification. Section X's structural finding – a majority of this dataset's items have their actual false claim living outside the specific post's own content, consistent with genuine footage recirculated under a new caption – is, on this evidence, a larger ceiling on end-to-end accuracy than any LLM-reasoning-focused fix. TruthLens has no reverse-image/video-search capability and has never claimed one.
4) Test the new validator check against phrasings it was not calibrated on. A genuine test of the label/reasoning-consistency check's generalization needs new cases this check was never looked at while being written (Section IX's disclosed circularity) – a concrete, scoped, named item.
5) Complete items 0008/0009's full-TruthLens condition and re-verify the vision-context fix end-to-end, both blocked by the same Gemini free-tier quota exhaustion (Section VII), once quota resets.
6) A dedicated matched-pair sourcing pass for RQ5. Bias evaluation needs items sourced for topic/structure/difficulty match across political actors, not actor diversity – a different sourcing objective than this program has pursued (Section XI).
7) Calibration, once the dataset is large enough to bin. Section XI's pre-registered gate (no confidence bin under 3 items) was not met at $n = 9$; revisit once it is.
8) A second, independent annotator, to compute a real inter-annotator agreement statistic on Tier-2 items and on the draft judgments underlying Section IX and X (Section VI).
9) The .gov.in full-page-fetch reliability gap (Section IX-J) – 403s and TLS certificate failures specific to this domain category – remains open; closing it needs either a deliberately scoped certificate-verification exception or further diagnosis of the specific failures, not attempted in this program.
10) Near-duplicate OCR-frame-variant claim deduplication (Section X), a concrete, scoped, and as yet unfixed gap.

XVI. Conclusion

This paper reports the first evaluation of TruthLens against independent, held-out ground truth, and we lead with the least flattering result it produced – not a weakness in TruthLens, but in our own experimental design: a skeptical post-hoc audit found that our baselines had been given a cleaner claim input than TruthLens's own extraction ever produced, across this entire program, contradicting our own stated methodology and inflating baseline performance. We report this first because our own protocol requires it, and because a research program that only audits the system under test, never its own baselines, is not applying the standard this paper argues for uniformly. Corrected – baselines re-run per real extracted claim, aggregated with TruthLens's own deterministic rule – full TruthLens beats two of three baselines outright and trails the third by a real but substantially smaller margin than the uncorrected comparison showed, on the same paired 6-item sample. The remaining gap traces to claim-extraction coverage failures on 2 items and one validator false negative, both independently diagnosed rather than inferred – more scientifically useful than either the original, partly-artifactual negative number or an uncritical positive one would have been. Set against this, a claim-decomposition counterfactual analysis shows the same architecture's aggregation mechanism delivering a real accuracy gain (50.0% vs. 25.0%, $n = 4$) once claim extraction has more than one claim to work with, localizing where this design's advantage does and does not currently materialize. We also show that two general, ground-truth-independent validator fixes measurably improved validator recall (16.7%→40%, 100% precision throughout) while leaving reel-level accuracy completely unchanged, both before and after the baseline correction – a concrete demonstration that verdict-label accuracy and the formally-defined support validity construct (Section IV-D) are separable properties of a fact-checking system, and, we argue, the more important axis for an architecture whose entire premise is evidence-before-verdict rather than label-matching. We report the evidence-quality analysis behind these results in full, including a gap between finding topically-relevant primary sources (68.75%) and extracting usable evidence from them (18.75%) that we consider a more actionable diagnosis than either number alone, a multimodal claim-coverage result in which OCR-derived evidence recovered a claim audio transcription missed entirely, and a structural finding – most of this dataset's actual false claims live outside the specific post being fact-checked – that we believe sets a real ceiling on single-post claim extraction regardless of how much its component quality improves. We report RQ5 (bias) and RQ6 (calibration) as deferred, not forced, because this dataset's size cannot yet support either, consistent with the discipline we tried to hold throughout this program: report what the data shows, including when that is a negative result, a deferred question, a bug in our own baselines, or a threat to validity in our own fix. The taxonomy of concrete failure modes in Section XII – 21 entries, over a third of them found and fixed during this evaluation program itself, each with a deterministic check and a regression test built from the real case that surfaced it – is, we believe, as practically useful a contribution as any single number in the tables above, because most of these failures, including the baseline confound this conclusion opens with, are only discoverable by running a complete system, and a complete audit of one's own comparisons, against real content, not by unit-testing components in isolation. Scaling the held-out benchmark past $n = 9$, closing the claim-extraction coverage gap this evaluation diagnosed, and running the matched-pair and calibration analyses this paper explicitly withholds are the concrete next steps, not vague future work, and are where we intend to take this program next.

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